INCH-POUND

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October 28, 1994

COMMERCIAL ITEM DESCRIPTION

GENERATOR, PULSE

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all agencies

1. SCOPE

This CID describes a pulse generator with a delay range from 0 to 99 99 milliseconds This CID is meant as a minimum requirement for the pulse generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID

2. CLASSIFICATION

Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description The equipment shall be Type III, Class 5, Style E and Color R in accordance with MIL-T-28800

3. SALIENT CHARACTERISTICS

3 1 Design and construction The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the Type III safety requirements of MIL-T-28800

3 2 Electrical power sources The equipment shall operate from nominal commercial, military and shipboard power sources of 115/230 volts AC (±10%) at line frequencies of 50 to 60 Hz Transient state conditions shall meet MIL-T-28800 Class 5 requirements

Beneficial comments, recommendations, additions, deletions, clarifications, etc , and any data which may improve this document should be sent to The Resources and Logistics Services Division, SA-ALC/TILDD, Bldg 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6524

AMSC N/A FSC 6625

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3 3 Temperature requirements

(a) Operating temperature The equipment shall meet its performance and accuracy requirements in an operating environment of +5°C to +55°C

(b) Non-operating temperature The equipment shall meet its performance and accuracy requirements in a non-operating environment of 0°C to +60°C

3 4 Mean time between failures (MTBF) The equipment MTBF shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours

3 5 Pre-conditioning requirement Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase

3 6 Calibration and maintenance adjustments The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours

3 7 Performance characteristics

(a)Pulse delay

(1)Delay range The delay range of the generator shall be from 0 to 99 99 milliseconds (ms), selectable in 1 nanoseconds (ns) increments

(2)Delay accuracy The minimum delay accuracy shall be ± (1 ns + time setting x time base stability)

(b)Pulse width

(1)Pulse width range The pulse width range of the generator shall be from 5 to 9 99 ms, selectable in 1 nanoseconds increments

(2)Pulse width accuracy The minimum pulse width accuracy shall be ±1 ns plus time base stability)

(c)Time base stability

(1)Long term The long term stability shall be better than $3x10^{-9}$/day.

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(2)Temperature The temperature stability shall be better than $1x10^{-8}$°C.

(d)Insertion delay A delay between an external trigger and an initial pulse shall not exceed 150 ns

(e)Jitter A jitter between an external trigger or an initial delay and the output pulse shall not exceed 250 picoseconds rms

(f)Trigger The trigger modes shall be manual, internal and external and be front panel selectable

(g)External trigger The pulse generator shall start to trigger when an input pulse, which has a minimum amplitude of 200 millivolts or less and a minimum pulse width of 10 ns or less, is applied The trigger shall be -2 to +2 volts and have a positive, negative slope selection switch

(h)Repetition rate The maximum repetition rate of the pulse generator shall be at least 5 MHz

(i)Initial and delay pulse output The pulse generator output shall be 5 volts into 50 ohms, the rise and fall time shall be less than 5 ns, selectable complements

(j)Internal clock modes The clock shall operate from an internal oscillator that has a frequency of 5 or 10 KHz

(k)Remote operation The generator shall be capable of remote operation with IEEE 488 interface bus All front panel functions and controls except power, manual trigger and polarity shall be able to be initiated remotely via IEEE 488 bus

4. REGULATORY REQUIREMENTS

In accordance with Section 23 403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials

5. PACKAGING AND MARKING

The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage The shipping shall be in compliance with national motor freight classification and uniform freight classification Shipping container shall be marked as specified in the contract or order

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6. QUALITY ASSURANCE

Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID, and that the product conforms to the producer's own drawings specifications standards and quality assurance practices

The Government reserves the right to require proof of such conformance prior to first delivery, and thereafter, as may be otherwise provided for under the provisions of the contract Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract)

7. NOTES

Metric products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376, and all other requirements of this CID are met

Federal and Military specifications are available from the Defense Documents Order Desk Building 4D, 700 Robbins Avenue Philadelphia, PA 19111-5094

Known acceptable products The following products are known to have previously met the Salient Characteristics of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification

Manufacturer	Model
Berkeley Nucleonic Corp	7085
Hewlett Packard	5359A

MILITARY INTERESTS

Preparing activity
Air Force - 82

Custodian.
Air Force - 99

(Project No 6625-0805)

Reviewing Activites.
Navy - AS
DLA - ES

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